

Arranque:

```
tom@MacBook-Neo-de-Tomy ~ % cd fixture2030-iris-objetos
tom@MacBook-Neo-de-Tomy fixture2030-iris-objetos % chmod +x scripts/*.sh
tom@MacBook-Neo-de-Tomy fixture2030-iris-objetos % bash scripts/01_inicializar.sh
[+] up 11/11
  Image intersystems/iris-community:latest-cd Pulled      25.0s
  ✓ Network fixture2030-iris-objetos_default Created      0.1s
  ✓ Container fixture2030-iris Started                    1.8s
NAME          IMAGE          COMMAND          SERVICE    CREATED    STATUS    PORTS
fixture2030-iris intersystems/iris-community:latest-cd "/tini -- /iris-main" iris        3 seconds ago Up 1 second (health: starting) 127.0.0.1:1972->1972/tcp, 127.0.0.1:52773->52773/tcp
tom@MacBook-Neo-de-Tomy fixture2030-iris-objetos % docker compose ps
NAME          IMAGE          COMMAND          SERVICE    CREATED    STATUS    PORTS
fixture2030-iris intersystems/iris-community:latest-cd "/tini -- /iris-main" iris        29 seconds ago Up 28 seconds (healthy) 127.0.0.1:1972->1972/tcp, 127.0.0.1:52773->52773/tcp
tom@MacBook-Neo-de-Tomy fixture2030-iris-objetos % docker compose logs --tail=40 iris
fixture2030-iris | 10/09/26-20:09:01:523 (1716) 0 [Database.MountedRW] Mounted database /durable/mgr/irismetrics/ (SFN 6) read-write.
fixture2030-iris | 10/09/26-20:09:01:525 (1716) 0 [Utility.Event] Processing Namespaces section
fixture2030-iris | 10/09/26-20:09:01:526 (1716) 0 [Utility.Event] Loading all namespaces
fixture2030-iris | 10/09/26-20:09:01:558 (1664) 1 [Generic.Event] Cannot lock /usr/irisys/mgr/enslib/ err(13): will try accessing readonly
fixture2030-iris | 10/09/26-20:09:01:560 (1716) 0 [Database.MountedRO] Mounted database /usr/irisys/mgr/enslib/ (SFN 7) read-only. File or filesystem allows read-only access.
fixture2030-iris | 10/09/26-20:09:01:700 (1716) 0 [Utility.Event] Activating namespaces
fixture2030-iris | 10/09/26-20:09:01:703 (1716) 0 [Utility.Event] Activating new namespace map
fixture2030-iris | 10/09/26-20:09:01:729 (1716) 0 [Utility.Event] Namespace changes have been activated
fixture2030-iris | 10/09/26-20:09:01:813 (1716) 0 [Utility.Event] Network Lock Upload Phase Starting
fixture2030-iris | 10/09/26-20:09:01:814 (1716) 0 [Utility.Event] Lock Upload Phase Complete
fixture2030-iris | 10/09/26-20:09:01:814 (1725) 0 [Utility.Event] Starting superserver on port 1972
fixture2030-iris | 10/09/26-20:09:01:817 (1716) 0 [Utility.Event] Processing Miscellaneous section
fixture2030-iris | 10/09/26-20:09:02:191 (1734) 0 [Utility.Event] Starting Servers
fixture2030-iris | 10/09/26-20:09:02:200 (1716) 0 [Utility.Event] Processing Devices section
fixture2030-iris | 10/09/26-20:09:02:202 (1716) 0 [Utility.Event] Processing DeviceSubTypes section
fixture2030-iris | 10/09/26-20:09:02:203 (1716) 0 [Utility.Event] Processing MagTape section
fixture2030-iris | 10/09/26-20:09:02:207 (1716) 0 [Utility.Event] Processing IO section
fixture2030-iris | 10/09/26-20:09:02:207 (1716) 0 [Utility.Event] Processing SQL section
fixture2030-iris | 10/09/26-20:09:02:211 (1716) 0 [Utility.Event] Processing Macro Pre-Processor (MPP) section
fixture2030-iris | 10/09/26-20:09:02:216 (1716) 0 [Utility.Event] Processing OS Package Requirements section
fixture2030-iris | 10/09/26-20:09:02:423 (1716) 0 [Generic.Event] Auditing to /durable/mgr/irisaudit/
fixture2030-iris | 10/09/26-20:09:02:535 (1716) 0 [Utility.Event] Rebuilding Extent index for namespace USER
fixture2030-iris | 10/09/26-20:09:02:542 (1716) 0 [Database.MountedRW] Mounted database /durable/mgr/user/ (SFN 8) read-write.
fixture2030-iris | 10/09/26-20:09:02:630 (1734) 0 [Utility.Event] Private webserver started on 52773
fixture2030-iris | 10/09/26-20:09:02:708 (1734) 0 [Utility.Event] Processing Shadows section (this system as shadow)
fixture2030-iris | 10/09/26-20:09:02:741 (1734) 0 [Utility.Event] Processing Monitor section
fixture2030-iris | 10/09/26-20:09:03:212 (1812) 0 [Utility.Event] WorkQueue: Starting work queue daemon parent=1734
fixture2030-iris | 10/09/26-20:09:03:848 (1734) 0 [Utility.Event] Processing Logging section
fixture2030-iris | 10/09/26-20:09:03:856 (1734) 0 [Utility.Event] Shard license: 0
fixture2030-iris | 10/09/26-20:09:03:861 (1898) 0 [Utility.Event] [SYSTEM MONITOR] System Monitor started in %SYS
fixture2030-iris | 10/09/26-20:09:05:958 (1715) 0 [Database.StartExpansion] Starting Expansion for database /durable/mgr/user/. 10 MB requested.
fixture2030-iris | 10/09/26-20:09:05:969 (1715) 0 [Database.FullExpansion] Expansion completed for database /durable/mgr/user/. Expanded by 10 MB. New size is 11 MB.
fixture2030-iris | 10/09/26-20:09:05:971 (1716) 0 [Utility.Event] Rebuild of Extent Index completed
fixture2030-iris | 10/09/26-20:09:05:981 (1716) 0 [Utility.Event] Enabling logons
fixture2030-iris | 10/09/26-20:09:05:986 (1716) 0 [Utility.Event] Interoperability System including auto-start of productions Starting
fixture2030-iris | 10/09/26-20:09:05:993 (1716) 0 [Utility.Event] Initializing Interoperability during system startup
fixture2030-iris | 10/09/26-20:09:06:070 (1897) 0 [Utility.Event] Starting TASKMGR
fixture2030-iris | 10/09/26-20:09:06:631 (1716) 0 [Utility.Event] Interoperability System Startup Complete
fixture2030-iris | 10/09/26-20:09:06:636 (1716) 0 [Utility.Event] System Info: $system=cdb460d3da3:IRIS GUID=45468FE6-C41D-11F1-8FB4-D618F0934053 SystemMode=
fixture2030-iris | [INFO] ...started InterSystems IRIS instance IRIS
tom@MacBook-Neo-de-Tomy fixture2030-iris-objetos %
```

Compilacion de clases:

```
tom@MacBook-Neo-de-Tomy fixture2030-iris-objetos % bash scripts/02_compilar_clases.sh

Node: cdeb460d3da3, Instance: IRIS

USER>

USER>

USER>

=== HITO 9: importando clases Fixture ===

USER>

USER>

Load started on 10/09/2026 20:18:11
Loading file /clases/Fixture.Persona.cls as udl
Load finished successfully.

USER>
Load Fixture.Persona; error? 0

USER>

Load started on 10/09/2026 20:18:11
Loading file /clases/Fixture.Jugador.cls as udl
Load finished successfully.

USER>
Load Fixture.Jugador; error? 0

USER>

Load started on 10/09/2026 20:18:12
Loading file /clases/Fixture.Arbitro.cls as udl
Load finished successfully.

USER>
Load Fixture.Arbitro; error? 0

USER>

Load started on 10/09/2026 20:18:12
Loading file /clases/Fixture.Tecnico.cls as udl
Load finished successfully.

USER>
Load Fixture.Tecnico; error? 0

USER>
```

```
Load started on 10/09/2026 20:18:12
Loading file /clases/Fixture.Partido.cls as udl
Load finished successfully.

USER>
Load Fixture.Partido; error? 0

USER>

Load started on 10/09/2026 20:18:12
Loading file /clases/Fixture.Evento.cls as udl
Load finished successfully.

USER>
Load Fixture.Evento; error? 0

USER>

Load started on 10/09/2026 20:18:12
Loading file /clases/Fixture.Demo.cls as udl
Load finished successfully.

USER>
Load Fixture.Demo; error? 0

USER>

=== Compilando todas las clases Fixture ===

USER>

Compilation started on 10/09/2026 20:18:12 with qualifiers 'ck'
Compiling 7 classes
Compiling class Fixture.Demo
Compiling class Fixture.Partido
Compiling class Fixture.Persona
Compiling class Fixture.Evento
Compiling class Fixture.Arbitro
Compiling class Fixture.Jugador
Compiling class Fixture.Tecnico
Compiling table Fixture.Partido
Compiling table Fixture.Arbitro
Compiling table Fixture.Jugador
Compiling table Fixture.Evento
Compiling table Fixture.Tecnico
Compiling table Fixture.Persona
Compiling routine Fixture.Demo.1
Compiling routine Fixture.Persona.1
Compiling routine Fixture.Partido.1
Compiling routine Fixture.Tecnico.1
Compiling routine Fixture.Jugador.1
Compiling routine Fixture.Evento.1
```

USER>

USER>

USER>

COMPILACION OK: Fixture.*

USER>

tomy@MacBook-Neo-de-Tomy fixture2030-iris-objetos %

Demostracion de objetos:

```
scripts > cargar_clases.mac
1 ; Carga definiciones fuente sin compilar individualment
2 ; completo para que IRIS vea simultaneamente los dos ex
3 Write !!, "=== HITO 9: importando clases Fixture ===", !
4 Set err = ""
5 Set sc = $System.OBJ.Load("/clases/Fixture.Persona.cls"

tomy@MacBook-Neo-de-Tomy fixture2030-iris-objetos % bash scripts/03_demo_objetos.sh

HERENCIA OK: Jugador persistido; ID=1, nombre=Jugador de demostracion, posicion=Delantero

HERENCIA OK: Arbitro persistido; ID=2, licencia=LIC-H9-67852-73238

VALIDACION OK: IRIS rechaza Arbitro sin Licencia obligatoria.

ERROR #5659: Property 'Fixture.Arbitro::Licencia(5@Fixture.Arbitro,ID=)' required
VALIDACION OK: se rechaza Programado -> Finalizado sin pasar por En juego.
Transicion valida: Programado -> En juego.
Evento agregado a la relacion en memoria; todavia no se guardo por separado.
GUARDADO ATOMICO OK: se invoco Partido.%Save() una sola vez para persistir el arbol inicial.
Partido ID=1 Codigo=H9-M-67852-73238 Estado=En juego
Evento ID=1||1 Tipo=Gol Minuto=12

NAVEGACION OBJETOS OK: H9-M-67852-73238 -> Gol minuto 12
Relacion inversa: el evento referencia al partido H9-M-67852-73238

VALIDACION OK: partido Finalizado rechaza un evento en minuto 0.

PROYECCION MULTIMODELO - SQL SOBRE Fixture.Partido
SQL Partido: ID=1 Codigo=H9-M-67852-73238 ARG-FRA Estado=Finalizado Se de=Estadio Monumental
```

```
VALIDACION OK: partido Finalizado rechaza un evento en minuto 0.

PROYECCION MULTIMODELO - SQL SOBRE Fixture.Partido
SQL Partido: ID=1 Codigo=H9-M-67852-73238 ARG-FRA Estado=Finalizado Se de=Estadio Monumental
PROYECCION MULTIMODELO - SQL SOBRE Fixture.Evento
SQL Evento: ID=1||1 Tipo=Gol Minuto=12 Descripcion=Gol de demostracion

FIN DEMOSTRACION HITO 9
Codigo de partido usado: H9-M-67852-73238
Conserva este codigo para localizar la fila en SQL.
```

Comprobar persistencia:


```
● tomy@MacBook-Neo-de-Tomy fixture2030-iris-objetos % docker exec -it fixture2030-iris iris session IRIS
Node: cdab460d3da3, Instance: IRIS

USER>Set stmt=##class(%SQL.Statement).%New()
USER>Set sc=stmt.%Prepare("SELECT COUNT(*) AS Total FROM Fixture.Partido")
USER>Set rs=stmt.%Execute()
USER>Do rs.%Next()

USER>Write "Partidos persistidos=",rs.%Get("Total"),!
Partidos persistidos=2

USER>Set stmt=##class(%SQL.Statement).%New()
USER>Set sc=stmt.%Prepare("SELECT COUNT(*) AS Total FROM Fixture.Evento")
USER>Set rs=stmt.%Execute()
USER>Do rs.%Next()

USER>Write "Eventos persistidos=",rs.%Get("Total"),!
Eventos persistidos=2

USER>Halt
```

Guardado y navegación de objetos:

```
USER>
tomy@MacBook-Neo-de-Tomy fixture2030-iris-objetos % chmod +x scripts/06_prueba_arbol_objetos.sh
set -o pipefail
|bash scripts/06_prueba_arbol_objetos.sh 2>&1 | tee docs/evidencia/rf6_rf7_arbol.txt
```

Node: cdab460d3da3, Instance: IRIS

USER>

=====

USER>
PRUEBA RF6 / RF7 - ARBOL DE OBJETOS Y NAVEGACION

USER>
=====

USER>

USER>

USER>

USER>

USER>

USER>

USER>

USER>

USER>

USER>

USER>

USER>

USER>

USER>

USER>

USER>

USER>

USER>

```
USER>
Antes de guardar: PartidoCodigo=H9-ARB-67852-77820

USER>
Antes de guardar: EventoTipo=Gol Minuto=17

USER>
Operacion de persistencia: una unica llamada a partido.%Save() sobre el padre.

USER>

USER>
Error al guardar (0 significa que no hay error): 0

USER>

USER>

USER>

USER>

USER>

USER>
ID Partido=3 | Codigo=H9-ARB-67852-77820

USER>
ID Evento=3||3 | Tipo=Gol | Minuto=17

USER>

USER>

USER>

USER>
NAVEGACION PADRE -> HIJO: H9-ARB-67852-77820 -> Gol minuto 17

USER>
NAVEGACION INVERSA HIJO -> PADRE: H9-ARB-67852-77820

USER>
RESULTADO: RF6 y RF7 demostrados mediante objetos, sin SQL para navegar.

USER>
```

Rechazo de una propiedad obligatoria:

```
tomy@MacBook-Neo-de-Tomy fixture2030-iris-objetos % chmod +x scripts/07_prueba_required_arbitro.sh  
[bash scripts/07_prueba_required_arbitro.sh 2>&1 | tee docs/evidencia/rf5_required.txt
```

Node: cdab460d3da3, Instance: IRIS

USER>

=====

USER>

PRUEBA RF5 - RECHAZO DE PROPIEDAD [Required]

USER>

=====

USER>

USER>

USER>

USER>

USER>

USER>

USER>

USER>

USER>

Objeto creado: Fixture.Arbitro

USER>

Nombre=Arbitro incompleto de prueba | Rol=Asistente

USER>

Licencia: OMITIDA intencionalmente (propiedad [Required]).

USER>

USER>

Error al guardar (1 significa que IRIS rechazo el objeto): 1

USER>

ERROR #5659: Property 'Fixture.Arbitro:Licencia(2@Fixture.Arbitro,ID=)' required

USER>

PRUEBA PASADA: IRIS rechazo el objeto por una propiedad obligatoria ausente.